

**E-COMMERCE CUSTOMER RETENTION STRATEGIES WITH RELIABLE CHURN
PREDICTION MACHINE LEARNING MODELS**

Submitted by

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1. Introduction

1.1 Background of the Study

Customer Churn, that is the customer disengagement with a company has been a major concern for industries. After all, companies especially those operating in competition-stiff areas like e-commerce have strategized on keeping their customers. This is because it is usually less expensive to retain a customer than to acquire a new one. This research shows that the expenditures involved in getting a new customer tends to be five times more than what it would take to simply keep an already existing one (Blattberg *et al.*, 2008). Therefore, it is imperative for the business to learn why their customers leave and how to proactively reduce the rate of customer churn to be able to stay viable in the years to come.

With the rise of the digital age, it is easy to observe the rapid growth of the e-commerce market, which has in turn raised the level of customers' aspirations. Most e-commerce businesses nowadays aim at ensuring their customer base does not diminish by providing a gratifying shopping experience. On the other hand, the sheer volume of data and interactions customers have within a business organisation makes it hard to manually forecast and detect the indicators of possible churn (Wisesa, Adriansyah and Khalaf, 2020). This is the role that machine learning and predictive analytics plays. Thanks to these technological advancements, any organization can now employ the vast information at its disposal to gauge the likelihood of certain clients churning and implement retention measures beforehand. Having such capacity that increases the precision of forecasting churn helps to shape the companies' marketing activities, turn attention to clients more effectively, and ultimately enhance satisfaction of customers.

There are many reasons why customers leave an e-commerce company and they can differ greatly depending on the type of business, how customers behave as well as the market. Factors that are widely found include duration of customer retention, purchase habits, means of settlement, and levels of satisfaction. For example, one would expect customers who have spent a shorter duration with the organisation or those who are unsatisfied with the services offered to the customers to defect. Likewise, those who shift to easier to use systems of paying or do not find the products that they are looking for in the stores are also predisposed to leave. These factors are very important for the reasons for churn prediction in cache - optimal for each concrete customer in the system (Shobana *et al.*, 2023).

Training such algorithms specifically for predictive analytics has been found to be useful in such contexts. These approaches can analyse past trends to find changes in customer behaviour that cyclically correlate with customer churn (Agrawal *et al.*, 2018). In contrast to classical statistical techniques, machine learning algorithms work with big data, model complex and non-linear dependencies, and recalibrate to reflect fresh data patterns and newly developed services (Çelik and Osmanoglu, 2019). This largely explains the ability to study customer behaviour in highly active and competitive industries such as e-commerce. In addition, over the years, the black box nature of some of the models has diminished which allows companies not only to forecast churn but also to appreciate the factors leading to a looming customer churn (Rudin, 2019).

Customer churn predictive modelling has seen quite an advancement in the last couple of years. At first, classical forecasting relied solely on statistical methods common for any market analysis. Those traditional approaches helped understand the chances of existing customers leaving, but could not process big data nor draw complex interdependencies between variables (Duchemin and

Matheus, 2021). However, with the expansion of these models to include more complex ones like decision trees, random forests, and gradient boosting models, churn models have become more precise and sturdier. Out of these, ensemble techniques, particularly XGBoost, have gained a lot of attention since they allow the mix of many non-performing models to build an efficient one. Aim of this project is to focus on building a predictive model for customer churn using machine learning techniques that will have both high accuracy and interpretability. This work uses data from an e-commerce site that contains many variables connected to users and their shopping habits as well as engagement with the website itself. With this in focus we will conduct an analysis of the data to figure out underlying trends that propel the churn rate and use them to build a suitable predictive model. The research undertakes measures to prepare a model that does not only offer accurate prediction but also contains a perceptible meaning that can in turn be meaningfully pursued by the business in question. The approach to research proposes to broaden the scope of understanding of churn in the context of e-commerce and its predictive models towards more useful outcomes for the business in terms of customer retention.

1.2 Research Aim and Objectives

The goal of this study is to create a reliable and understandable system aimed at estimating the customer churn rates of e-commerce businesses. To this end, the study seeks to accomplish the following tasks: to pinpoint the key elements in customer churn and the significance of each; to build a machine learning model that will be successful in predicting churn; to assess the performance of the model by employing a range of indicators, such as accuracy, precision, recall and F1 score; and to make recommendations to business management on how to tackle the problems of customer attrition and improvement of customer retention.

1.3 Research Questions

This study seeks to address the following research questions:

What are the key factors influencing customer churn in the e-commerce sector?

How accurately can a machine learning model predict the likelihood of customer churn?

Which machine learning algorithms provide the best balance between accuracy and interpretability for churn prediction?

How can the insights from the model be used to develop effective customer retention strategies?

2. Literature Review

2.1 Overview of Customer Churn

Lately and particularly with the recent rise of e-commerce where all the concerns levelled about customer retention are geared towards increasing the business's bottom line, prediction of customer churn has received a lot of focus and attention. The ability to predict churn allows organisations to implement specific customer retention plans, preventing the aforementioned losses. Further advancements on this topic have employed several machine learning (ML) techniques, but the results have had both achievements and significant challenges.

2.2 Importance of Churn Prediction in E-Commerce

2.2.1 Trends in Customer Behaviour

Nagesh (2023) emphasises the need to develop flexible models for churn prediction as traditional models are built on rigid, predetermined strategies. Such inflexibility limits the capacity of firms to adjust to variable consumer behaviour particularly in e-commerce markets, where preferences and choices are determined in a very short span of time. Therefore, research is focused on machine learning models. Such models have an advantage in that they do not have to follow any set rules

on how to analyse data and can learn patterns within it, thus being more welcoming to new customers.

2.2.2 Role of Retention in Business Strategy

In the context of the performance of the model, the term ‘overfitting’ is defined with specific and more elaborate explanations by Anuradha and Kalsi (2021) especially when dealing with unbalanced datasets during training. Conceived in its most basic sense, overfitting occurs when the model is too complex for the training data, producing good accuracy scores with the data used for training, while performing poorly with any generalised unseen data. This poses a great challenge in many churn forecasting practices especially since it is observed that the number of customers who churn is usually very low when compared to the number of customers who do not churn thus creating a class imbalance in the training data. Solutions to the problem of class imbalance exist for instance the use of the armouring method of over-sampling known as Synthetic Minority Over-sampling Technique SMOTE, however such techniques are not without consequences; for instance, the creation of new data points tends to exacerbate the problem of overfitting.

2.3 Machine Learning Techniques in Churn Prediction

2.3.1 Traditional Statistical Methods

Customer churn prediction is difficult, according to Wu (2022), primarily because of challenges associated with feature extraction. Sophisticated model building, however, necessitates identifying the appropriate reaction in regard to behaviour, who is active, and also the degree of attention paid. Still, ineffectively performed feature extraction will lead to an unsatisfactory performance of the model even in use of better techniques; for instance, in e-commerce, the types of features would

include the browsing tendencies of a customer before making a purchase, purchasing frequency and their exposure to verbal calls to action all which are important engagement metrics with customers among others.

2.3.2 Modern Machine Learning Approaches

Xiahou and Harada (2022) state that it is common to observe that a lot of models cannot be generalised across models. Churn prediction models can be said to have an application all over the world, but a model trained on retail and e-commerce will be of no use in B2B or finance-directed stratagems as there will be different consumer tendencies and churn causes. Thus, the application of the churn prediction models becomes limited which indicates the necessity for adapted churn prediction models aimed at certain sectors only and general approaches which are flexible enough to suit different sectors.

Supervised Machine Learning Models

Elgohary et al. (2023) address another important problem, especially with respect to deep learning models, which is the troubling and often very time consuming hyperparameters tuning. It is accustomed to modify the model, which usually takes a lot of time and resources, making it difficult to apply the model in real time situations. To overcome this limitation, methods driven by hyperparameter optimization algorithms have been developed and begin to be used, though still to a limited extent. In this paper we will use three supervised classification machine learning models which are described below.

Logistics Regression

Logistic Regression is a statistical approach which is primarily applied in binary classification problems, where the dependent variable can take one of the two states (for example "Yes" or "No", "0" or "1", "Success" or "Failure" etc.). Unlike linear regression, which estimates random values, logistic regression gives an estimate of the class probability. Hence it finds applications in many sectors such as medicine, marketing, social sciences to name a few as it helps in predicting the likelihood of a certain event occurring (Peng et al., 2002).

In Logistic Regression, a linear combination of the input features is formed, however where this output is actually predicted, it is transformed using the sigmoid function (sometimes called the logistic function) to squash it in the range of 0 and 1, thus representing probabilities. The sigmoid function, which is also called the logistic function, is given by:

$$\hat{y} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}}$$

where

$\hat{y}$ represents the predicted probability of the positive class (e.g., churn = 1)

e is the base of the natural logarithm

β_0 is the intercept terms

$\beta_1, \beta_2, \dots, \beta_n$ are the coefficients of the predictor variables $x_1, x_2, \dots, x_n$

$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$ is a linear expression, similar to what is used in linear regression (Hosmer & Lemeshow, 2000).

The sigmoid function has the shape of an "S" and takes the input values to a range of probabilities

between 0 and 1 that can thereafter be viewed as the likelihood of being in the positive class (James et al., 2013).

Decision Boundary: After the model delivers its probability prediction $\hat{y}$, the outcome must be classified into either of the two classes by employing a threshold. Usually, this threshold is specified at 0.5:

If $\hat{y} \geq 0.5$, the output is classified as "1" (positive class).

If $\hat{y} < 0.5$, the output is classified as "0" (negative class) (Field, 2009).

Parameter Estimation: The coefficients $\beta_1, \beta_2, \dots, \beta_n$ are estimated using a technique called Maximum Likelihood Estimation (MLE). In the MLE process, the ultimate goal is to estimate the parameters by maximising the probability of obtaining the given sample data. The parameter in the likelihood function expresses the probability of the occurrence of the observed data for a given set of parameters.

Logistic regression seeks to maximise the log of this likelihood function (log-likelihood), as it is computationally easier to handle (Menard, 2002).

Log-Likelihood Function:

The log-likelihood function for logistic regression is:

$$\text{Log - Likelihood} = \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Here, y_i represents the actual outcome (0 or 1) for each observation, and $\hat{y}_i$ represents the predicted probability of the positive class for the i^{th} observation. Maximising this function finds the optimal values for the coefficients, making the model best fit the data (Agresti, 2002).

Interpretation of Coefficients: Each coefficient β_j in logistic regression represents the effect of a predictor x_j on the log-odds of the outcome. The relationship between a predictor and the outcome is expressed in terms of odds ratio:

$$\text{Odds Ratio} = e^{\beta_j}$$

If β_j is positive, the odds of the event occurring increase with the predictor.

If β_j is negative, the odds decrease as the predictor increases (Allison, 2012).

Random Forest Classifier

The random forest algorithm is a type of ensemble learning which can be employed for both classification and regression tasks. It creates many decision trees and combines their predictions in order to produce the result. This method is quite useful for the real world for more complex purposes because it alleviates the problem of overfitting to some extent, which is a disadvantage in a single decision tree. Random Forest has been employed for different purposes across several fields such as finance, healthcare and image recognition, where it has proved to be accurate and reliable (Breiman, 2001).

Random Forest is a type of decision tree model which is built using a method called bootstrap aggregation or bagging whereby a number of decision trees are constructed from the training data. More specifically, training data is divided into multiple subsets, each of which is utilised to fit a particular decision tree by sampling the training data with replacement (Ho, 1995).

Feature selection is done with randomness: In the construction of each Decision Tree, besides other factors, Random Forest applies another level of randomness by selecting a few characteristics only for splitting the nodes. This aids in ensuring that the trees within the forest are not overly identical

to one another and helps to reduce the correlation of the trees, as well. Generally, in the case of classification problems, the number of features that are selected is specified as $\sqrt{p}$, where p denotes the total number of features. (Hastie, Tibshirani, & Friedman, 2009).

Every tree in the forest is tasked with predicting a class label for a particular instance, which is the input data point. The Random Forest classifier combines class labels in most instances by the process of tallied votes:

When most of the trees in the random forest predict the output to be Class A, the end result will also be Class A. In the contrary case, if most of the trees predicted Class B, then the output will be Class B as well (Breiman, 2001).

Estimation of Error Rate Based on Out-of-Bag Samples: As each of the trees is generated based on a bootstrap sample of the original dataset, a little less than one third of the dataset is untouched in the fitting of each of the individual trees. The samples that are not used for training are termed Out of Bag samples. Out of Bag samples are utilised by the Random Forest in the assessment of the error rate and the accuracy of the model without really subjecting the data to external validation (Liaw & Wiener, 2002).

Importance of Individual Features: According to literature, Random Forest comes with added benefits such as efficiency in estimating the importance of each of the predictors in the prediction. This would involve estimating how much the accuracy of the model or the Gini impurity of the model decreases if this variable is not considered, or is permutation tested. Accuracy decreases which follow considerable differences in the importance value are thus termed as more important features (Strobl et al., 2009).

Gini Impurity: In random forest, the Gini impurity is the general method employed in splitting the nodes in each decision tree. Gini Impurity refers to the chances that a randomly chosen member of the group would be labelled incorrectly, if the labelling was done at random according to the distribution of the labels in the group:

$$Gini(D) = 1 - \sum_{i=1}^c p_i^2$$

D is the dataset.

C is the total number of classes

p_i represents the proportion of instances in class i

Lower Gini values indicate purer nodes, which means a better split (Hastie, Tibshirani, & Friedman, 2009).

XGBoost Classifier

The term “XGBoost” stands for Xtreme Gradient Boosting, which is a high-level algorithm used for constructing effective predictive models, more specifically through the process of gradient boosting. Performing all other stated objectives, it is most suited for the cases where the task is classifying the data into two or more groups, e.g. whether the email is spam or not – whether there is a disease or not. XGBoost is praised for its short training times and high accuracy to the extent of being an industry and academia standard (Chen & Guestrin, 2016).

Gradient Boosting Framework: XGBoost is built upon the principle of gradient boosting, which refers to an effective method whereby several simple models (weak learners) are fitted in stages. Each such stage is built on the function of the previous stages and is meant to reduce the errors

produced by them. Decision trees are typically used in these models. The objective is to add a new model in order to reduce the errors or residuals (Friedman, 2001).

Additive Learning: The construction of trees is sequential whereby trees are added in order to enhance the overall prediction. For the new tree being added at step t :

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta \cdot f_t(x_i)$$

$\hat{y}_i^{(t)}$ is the updated prediction after adding the i^{th} tree.

η is the learning rate, which controls how much each tree influences the overall prediction.

$f_t(x_i)$ represents the output of the new tree.

Loss Function and Regularization: Acknowledging that XGBoost trains a model in the process of minimising the empirical risk (loss function) associated with the fit and incorporating the complexity of the model (regularisation component) so as not to overfit, that is, to achieve good results on the training data, while performing poorly on new data (Chen & Guestrin, 2016):

Objective Function = Loss Function + Regularization term

Here objective function should be minimised.

$$\text{Objective function} = \sum_{i=1}^n L(y_i, \hat{y}_i) + \sum_{t=1}^T \Omega(f_t)$$

$L(y_i, \hat{y}_i)$ measures how well the predictions match the actual outcomes.

$\Omega(f_t)$ penalises complex trees, keeping the model simpler

T represents the total number of trees in the ensemble.

Tree Pruning: Based on the training phase of XGBoost, tree pruning is another strategy that enhances the performance of the model by eliminating branches that are deemed not useful. There

exists a parameter named gamma (γ) that governs the conditions under which a split in the tree may be regarded as valuable:

In case a split does not contribute significantly to the tree, it is eliminated for the sake of keeping the tree clean.

This is said to aid in ensuring model accuracy and model complexity are not in conflict with each other (Chen & He, 2015).

Second-Order Approximation: A key feature of XGBoost is its use of *second-order Taylor approximation*, which takes into account both the gradient (slope) and the curvature (shape) of the loss function. This leads to more precise updates and faster convergence:

$$\text{Objective function} \approx \sum_{i=1}^n [g_i f_i(x_i) + \frac{1}{2} (h_i f_t^2(x_i))] + \Omega(f_t)$$

g_i is the gradient, representing the direction in which to adjust the model.

h_i is the curvature (The Hessian), which adjusts the step size for better accuracy

$\Omega(f_t)$ penalises complex trees, keeping the model simpler

This approximation allows XGBoost to fine-tune the model more effectively (Chen & Guestrin, 2016).

Ensemble Methods for Classifiers

Ensemble methods like Voting Classifiers and Stacking Classifiers - all belong to more advanced and fundamental techniques in machine learning. They increase predictive quality while lessening overfitting effects, increase reliability, give more leeway, and help manage skewed datasets better.

Nonetheless, these techniques combine the power of several models allowing practitioners to construct predictive systems that are more precise and dependable decision making tools in different applications.

Voting Classifiers:

The Voting Classifier is an ensemble method which improves classification by integrating several models. This strategy works best because the base classifiers benefit from each other's strengths (Dietterich, 2000).

The fundamental concept behind a Voting Classifier is that the models which form it are assumed to agree in the form of some unified prediction. The first stage of this process is to take a number of different classifiers, to train them on the same dataset, possibly applying different algorithms and/or hyperparameters for training each of the classifiers. After training, given an input, one can get predictions from every single classifier. There are usually two types of voting: hard voting and soft voting.

In hard voting, the output class label is chosen based on the class predicted by the largest number of classifiers. This is very simple and works well when the classes in question are nearly balanced. Soft voting on the other hand, considers the class probability distributions obtained from each of the classifiers before making the final decision. In this case, the classifier gives a probability for each of the classes and the one with the highest average probability is considered the predicted class. This technique may result in better performance especially where there is class imbalance as class distributions can be skewed, with some classes being small and others large, (Zhou, 2012).

Stacking Classifiers:

Diversity is the primary distinguishing feature of the Stacking Classifier, or stacked generalisation,

as it is an ensemble learning paradigm where several different classifier models are trained, and further, a classifier is trained on the predictions of the base classifiers (Wolpert, 1992). In other words, this methodology attempts to make different algorithms work toward the same goal through the use of different algorithms in a nested hierarchy.

As for the Stacking Classifier, it is composed of several base models or so-called level-0 models which are trained on the data set in the first place. These base models have different types, for instance, ‘decision trees’, ‘logistic regression’, and ‘support vector machines’, among other types. After the training of the base models is completed, they predict the outcomes for the same data set and those outcomes are gathered in order to be used as a feature set for a higher-level model (the meta-classifier or level-1 model).

The upper model level, the meta-classifier, is fitted to the outcomes of the base classifiers in such a way that it perfectly understands how these base classifiers can be best combined in order to achieve higher accuracy. The meta-classifier can be an arbitrary classification algorithm and its purpose is to model how the predictions of the base models are combined such that performance is the best possible. In this case, methods like cross-validation tend to be used because it is important that the meta-classifier does not fit on the training data and is able to work on new data (Zhou, 2012).

Supervised machine learning model’s evaluation metrics:

Confusion Matrix: A Confusion Matrix describes all the predictions made by the model in detail including the numbers of true positives, true negatives, false positives, and false negatives. In addition, it also helps in understanding the type of mistakes the model commits (Chicco, 2020)

		Actual Values	
		Positive	Negative
Predicted Values	Positive	True Positive(TP)	False Positive(FP)
	Negative	False Negative(FN)	True Negative(TN)

Fig 1: Confusion Matrix

Accuracy defines the rate of correct classifications relative to all instances and hence sounds quite intuitive as a metric for model performance. It comes particularly handy where the distribution of various classes is evenly spread (Kohavi & Provost, 1998).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Here, TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

Metrics like Precision, Recall, and F1-Score are particularly important for imbalanced datasets where one class significantly outnumbers the other (Powers, 2011).

Precision measures the proportion of positive predictions that are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

Recall (or Sensitivity) measures the proportion of actual positives that are correctly identified.

$$Recall = \frac{TP}{TP + FN}$$

F1-Score is the harmonic mean of precision and recall, balancing the two metrics:

$$F1 - Score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

2.4 Limitations of Existing Churn Prediction Models

2.4.1 Overfitting and Data Imbalance

Those models, especially the deep-learning based ones, designed for churn prediction are inevitably faced with one of the most issue daunting concerns, and this is how they are made interpretable, or the so-called black box problem, as noted by Elgohary et al. (2023); It is further explained that most fears will come into play on the action or even in the response to the probability predictions of the model, in cases where there is no proper understanding of what makes customers leave. Which brings us to the regrettable, yet the existing, ‘struggle’ in the field of research, focused on explainable artificial intelligence. The term ‘explainable artificial intelligence’ denotes the area of artificial intelligence which focuses on making machine learning models transparent and comprehensible. By doing so, XAI would build the confidence of the firms to the output and encourage the firms to engage in more sophisticated measures aimed at reducing churn.

Challenges of Imbalanced Datasets

Apart from these technical challenges, Karaarslan et al. (2024) note that most existing models barely take into account factors from outside the established customer base which are known to affect the customer churning rate, such as industry trends and economic factors in particular. This

limitation in turn indicates that the predictive model of churn may be quite naive and lead to erroneous conclusions about retention strategies.

2.4.2 Lack of Feature Engineering

There exists an extensive range of literature on customers who churn (or cease purchasing), however, there is a scarcity of literature that would help devise the comprehensive retention schemes suited to forecasting studies. It is possible, Andrew (2024) surmises, to ascertain those customers who are likely to leave the company and address them, but that is noting more than the first step and that the most valuable part is in the measures taken once such information is received, he states. Such measures, he goes on, could encompass employing precise marketing strategies, designing more appealing loyalty programs, or improving the quality of services offered to the vulnerable customer segment.

2.4.3 Interpretability Issues in Machine Learning Models

Despite the challenges, it can be claimed that such churn prediction models have significantly improved in the last few years, primarily because of the artificial intelligence and machine learning in which Faritha Banu et al. (2022) lay great emphasis on. They maintain that models should not only be predictive and precise but also useful in the avoidance of firm's churn. In other words, the salient characteristics of churn should be established, and practicable measures to counteract them like discounts, promotional offers, and enhanced service quality should also be suggested.

2.5 Strategies for Improving Churn Prediction

2.5.1 Feature Engineering and Selection

As outlined by Matuszelański and Kopczewska (2022), whose approach to predicting the churn in e-commerce is by far more spatial as it seeks to combine both geography and customer

demographics in a bid to appreciate customer behaviour. Their study maintains the position that customer turnover is not only limited to the transaction between the customer and the enterprise but incorporates so many other aspects that include the locality of the customers' economic resource. In this situation, it means that these factors would improve the accuracy of the churn modelling and also the retention strategies employed.

2.5.2 Hybrid Models for Better Accuracy

It should be noted that the progress of hybrid modelling has also been embraced as an attempt to overcome the challenges brought about by the application of a single algorithm. Anuradha and Kalsi (2021) combine K-Means clustering and NLSVMs and argue that one can take advantage of the different algorithms for the improvement of the model. The disadvantage of hybrid models of this type is that they improve the accuracy of the model but at the expense of increasing computational burden and reducing interpretability of the model.

2.5.3 Explainable AI in Churn Prediction

While the current techniques are successful in quantifying the customer retention rate in a business, very often they are not helpful in suggesting the actual measures to retain the customers. This gap needs to be filled in particular, as it enhances the practical relevance of churn prediction.

2.6 Addressing the Gap in Actionable Retention Strategies

It's true that the models available to date work very well when estimating the customer attrition rate, yet they rarely include viable approaches to retaining the existing clientele. Filling this void is vital in increasing the scope of use of predicting customer churn. For that reason, the purpose of this study is to design a model for forecasting churn that deals with customer churn, but also finds useful retention measures based on data analysis. As noted by Xiao (2024) and Faritha Banu et al.

(2022), these measures can be incorporated into the churn forecasting process so that the at-risk customers are identified and steps taken to enhance their loyalty as well as reduce churn.

3. Methodology

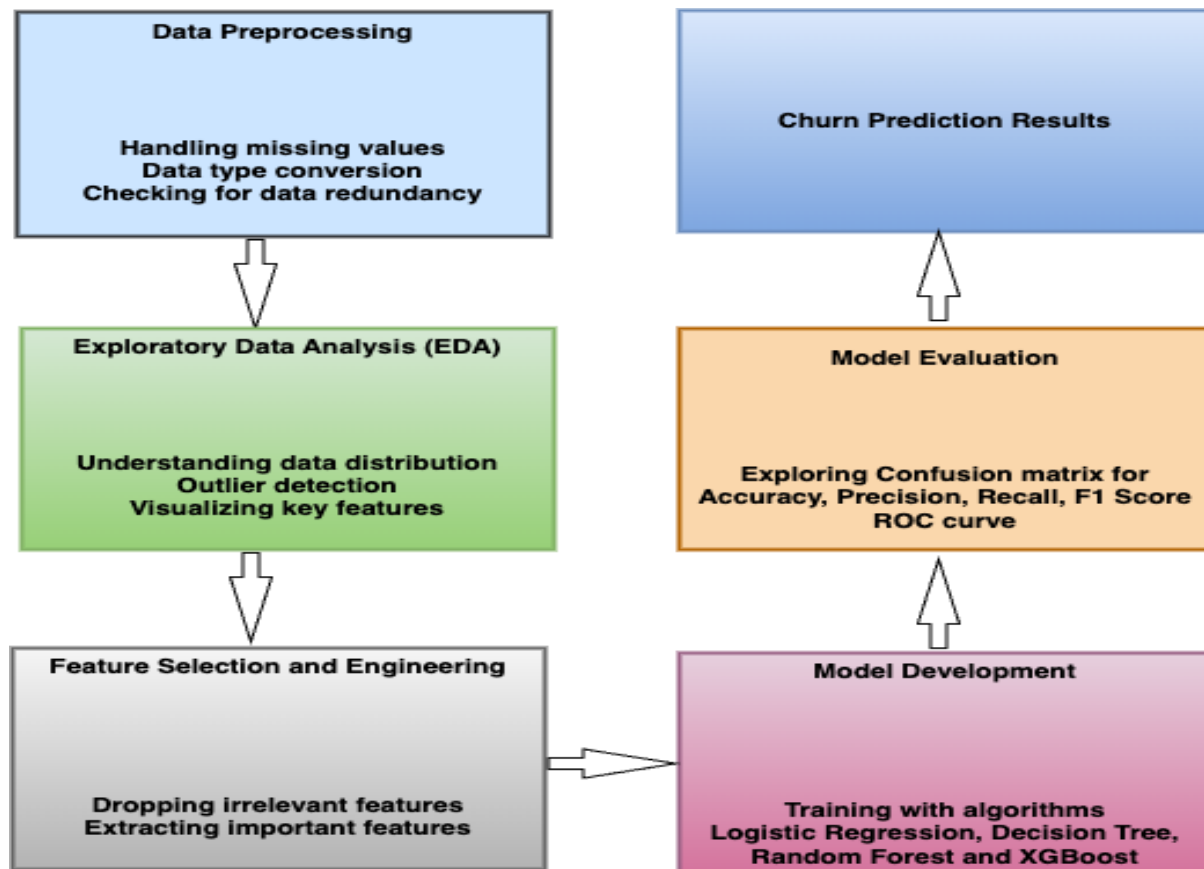


Fig 2: Methodology flow chart

3.1 Research Design

This study employs a data-centric method in predicting customer churn which aims at using machine learning features. Data pre-processing, Grasp and visualise the patterns in the data,

Construct the predictive models and test the models are the primary aims of the research. The existing machine learning models will be used to enable organisations to appreciate customer consumption patterns and more so to spot the customers who are likely to leave the organisation.

3.2 Dataset

The dataset uploaded by (Verma Ankit , 2010) is used in this study sourced from **Kaggle**, a platform providing publicly available datasets for machine learning projects. The dataset includes multiple features capturing customer demographics, behaviour, and transactions, focusing on churn prediction.

This dataset captures essential details about customer demographics, behaviours, and interactions with the e-commerce platform. These features are crucial for understanding the drivers behind customer churn. Below is an overview of variables in the dataset:

Data	Variable	Description
0 E Comm	Churn	Churn Flag
1 E Comm	Tenure	Tenure of customer in organization
2 E Comm	PreferredLoginDevice	Preferred login device of customer
3 E Comm	CityTier	City tier
4 E Comm	WarehouseToHome	Distance in between warehouse to home of customer
5 E Comm	PreferredPaymentMode	Preferred payment method of customer
6 E Comm	Gender	Gender of customer
7 E Comm	HourSpendOnApp	Number of hours spend on mobile application or website
8 E Comm	NumberOfDeviceRegistered	Total number of deceives is registered on particular customer
9 E Comm	PreferedOrderCat	Preferred order category of customer in last month
10 E Comm	SatisfactionScore	Satisfactory score of customer on service
11 E Comm	MaritalStatus	Marital status of customer
12 E Comm	NumberOfAddress	Total number of added added on particular customer
13 E Comm	Complain	Any complaint has been raised in last month
14 E Comm	OrderAmountHikeFromLastYear	Percentage increases in order from last year
15 E Comm	CouponUsed	Total number of coupon has been used in last month
16 E Comm	OrderCount	Total number of orders has been places in last month
17 E Comm	DaySinceLastOrder	Day Since last order by customer
18 E Comm	CashbackAmount	Average cashback in last month

Fig 3: Data Description

Preview of the dataset:

CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2
50002	1		Phone	1	8	UPI	Male	3	4	Mobile	3
50003	1		Phone	1	30	Debit Card	Male	2	4	Mobile	3
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5
50005	1	0	Phone	1	12	CC	Male		3	Mobile	5
50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone	5
50007	1		Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Accessory	2
50008	1		Phone	1	6	CC	Male	3	3	Mobile	2
50009	1	13	Phone	3	9	E wallet	Male		4	Mobile	3
50010	1		Phone	1	31	Debit Card	Male	2	5	Mobile	3
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others	3
50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion	3
50013	1	0	Phone	1	11	COD	Male	2	3	Mobile	3
50014	1	0	Phone	1	15	CC	Male	3	4	Mobile	3
50015	1	9	Mobile Phone	3	15	Credit Card	Male	3	4	Fashion	2
50016	1		Phone	2	12	UPI	Male	3	3	Mobile	5
50017	1	0	Computer	1	12	Debit Card	Female		4	Mobile	2
50018	1	0	Mobile Phone	3	11	E wallet	Male	2	4	Laptop & Accessory	3
50019	1	0	Computer	1	13	Debit Card	Male	3	5	Laptop & Accessory	3
50020	1	19	Mobile Phone	1	20	Debit Card	Female	3	3	Mobile Phone	4
50021	1	0	Mobile Phone	3	12	Debit Card	Male	3	5	Laptop & Accessory	3
50022	1	20	Mobile Phone	1	29	Credit Card	Female	3	3	Fashion	2

Fig 4: First preview of dataset

MaritalStatus	NumberOfAddress	Complain	OrderAmountHikeFromlastYear	CouponUsed	OrderCount	DaySinceLastOrder	CashbackAmount
Single	9	1	11	1	1	5	160
Single	7	1	15	0	1	0	120
Single	6	1	14	0	1	3	120
Single	8	0	23	0	1	3	130
Single	3	0	11	1	1	3	130
Single	2	1	22	4	6	7	130
Divorced	4	0	14	0	1	0	120
Divorced	3	1	16	2	2	0	120
Divorced	2	1	14	0	1	2	120
Single	2	0	12	1	1	1	120
Divorced	2	0		9	15	8	290
Single	10	1	13	0	1	0	150
Single	2	1	13	2	2	2	130
Divorced	1	1	17	0	1	0	130
Single	2	0	16	0	4	7	190
Married	5	1	22	1	1	2	120
Single	2	1	18	1	1	0	120
Single	2	1	11	1	1	3	150
Single	2	1	24	1	1	6	160
Divorced	10	1	18	1	4	3	150
Divorced	5	1	18	6	7	7	160
Divorced	2	0	12	11	15	6	200

Fig 5: Second preview of dataset

3.3 Data Preprocessing Techniques

The preprocessing phase focused on the aspect of data readiness for modelling. The dataset has 5630 rows and 20 columns of different data types with int64 representing integers, float64 representing decimals and object representing text variables. For most of the column's data is available completely, however for some other columns like Tenure and OrderCount data is missing and imputation techniques have to be applied.

The scaling was performed only and only on number features, to have their mean at 0 and standard deviation as 1. This is important for models that are affected by the scale of features. StandardScaler was the method used in this process, its main objective to ensure features range

varies, thus all numerical features are optimally used in the performance of the model.

Additionally, the dataset consists of qualitative features like PreferredLoginDevice, PreferredPaymentMode, Gender, PreferredOrderCat and MaritalStatus. These qualitative features were required to be encoded using one hot encoding which converts the said features to the numerical format suitable for machine learning models. This involved adding a separate column of dummy variables for each of the categories such that the algorithms can apply the grouping information appropriately without any confusion.

The class imbalance for the Churn variable included a disproportionate representation of approximately: 83.16% of instances were non-churners while 16.84% were churners.

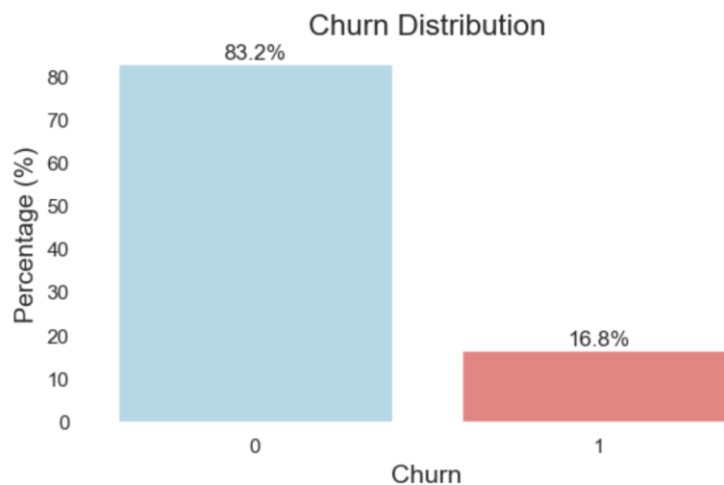


Fig 6: Percentage Distribution of Non-churn(0) and Churn(1) in the dataset

This imbalance was mainly resolved by the use of Synthetic Minority Over-sampling Technique (SMOTE). SMOTE is a technique that adjusts the dataset by creating new instances of the small class (Chawla et al., 2002). This change promotes the predictions' performance since it prevents

the models from overfitting only on the majority class and enables them to learn from all data, including unseen data.

3.4 Exploratory Data Analysis (EDA)

This included looking for trends and outliers by examining the distributions, central tendencies, and variabilities within the data set. Most statistical features were measured using means, medians, as well as standard deviations, whereby it was found that the SatisfactionScore had values underlying 3.07 while the OrderCount averaged about 3.01. It was also established that none of the features were normally distributed in a bell-shaped manner and this was exposed to risk of skewness and outliers in the data (NIST/SEMATECH, 2012).

The most extreme of all these HourSpendOnApp was found to be 127, showing that there are customers who tend to use the application to an excessive extent over its normal expectation. Such variations from the expected norm could represent outliers that may impact the analysis of the study. Churn, which was the dependent variable, had a lot of data deemed to be unrealistic as there were a majority of non-churners. In this case, this differential of Churn's segments averaged with extreme data during modelling of the variable forced application of sampling techniques (Chawla et al., 2002).

3.4.1 Handling Missing Data

As may be expected, one of the key aspects in preparing a dataset for analysis involves dealing with the absence of data. CustomerID, Churn, and Gender are among the few complete columns that have no missing values, while some columns such as Tenure, WarehouseToHome, and DaySinceLastOrder have missing values.

```
Tenure                4.69
WarehouseToHome      4.46
HourSpendOnApp        4.53
OrderAmountHikeFromLastYear 4.71
CouponUsed            4.55
OrderCount            4.58
DaySinceLastOrder     5.45
dtype: float64
```

To be exact, Tenure records 264 missing values (4.69%), WarehouseToHome 251 missing values (4.46%), and DaySinceLastOrder 307 missing entries (5.45%). It is important to resolve these issues since they may cause delays in statistical measures and model forecasts.

In view of the fact that most features do not adhere to a normal distribution, the

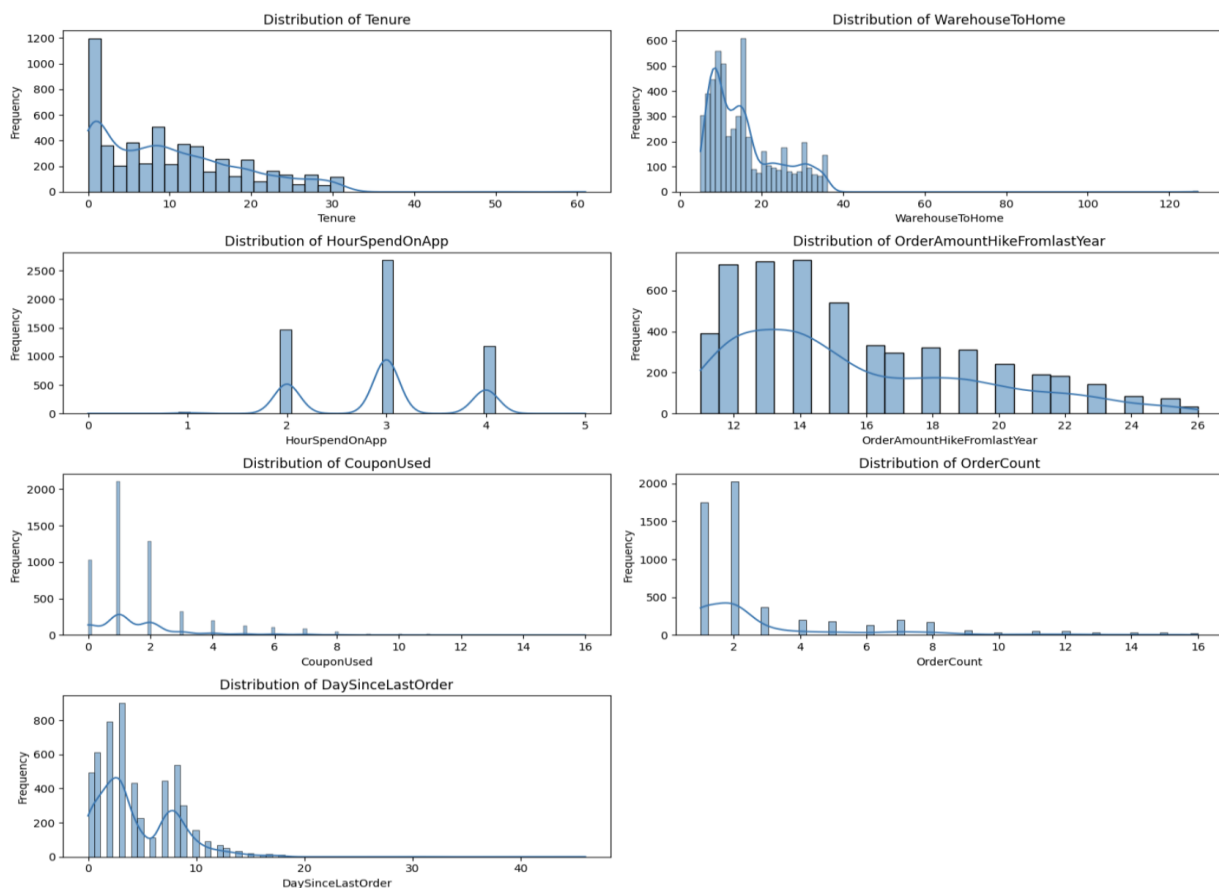


Fig 7: Histogram plot of features with missing values

median was adopted as the missing value imputation technique since it is less sensitive to extreme values (Rubin, 2004). This technique helps in keeping the original shape of the distribution of the data while also making sure that there are no incomplete data entries. Bias results because of data omissions during the analysis if proper strategies are not put in place to address missing data. Additionally, such data would make the prediction models less reliable.

3.4.2 Outlier Detection

We employed the Interquartile Range (IQR) Method. This technique entails deriving the lower and upper regions, which are also referred to as quartiles, of a particular numerical column. The respective quartiles represent the 25th and the 75th positions of the data as ordered. The term Interquartile Range is used to describe the distance from Q1 to Q3.

In order to find out the limits of the outliers, $Q1 - 1.5 * IQR$ is taken as the lower limit and $Q3 + 1.5 * IQR$ is the upper limit. Data points that are lower than the lower limit or higher than the upper limit are treated as outliers.

In the outlier detection phase of the dataset exploration, some features such as OrderCount, CouponUsed, CashbackAmount, and NumberOfDeviceRegistered stood out with a relatively higher number of outliers. Consequently, these features were chosen for outlier treatment to enhance data quality. In this analysis, the Churn column (which consists of 0 and 1 values only) was also presented. However, since this is a binary classification variable indicating two different outcomes, both lower and upper limits set for outlier detection were 0, meaning that every observation could theoretically be classed as an outlier. It must be emphasised, however, that those binary values are the core of the classification task and should not be considered outliers in this

scenario.

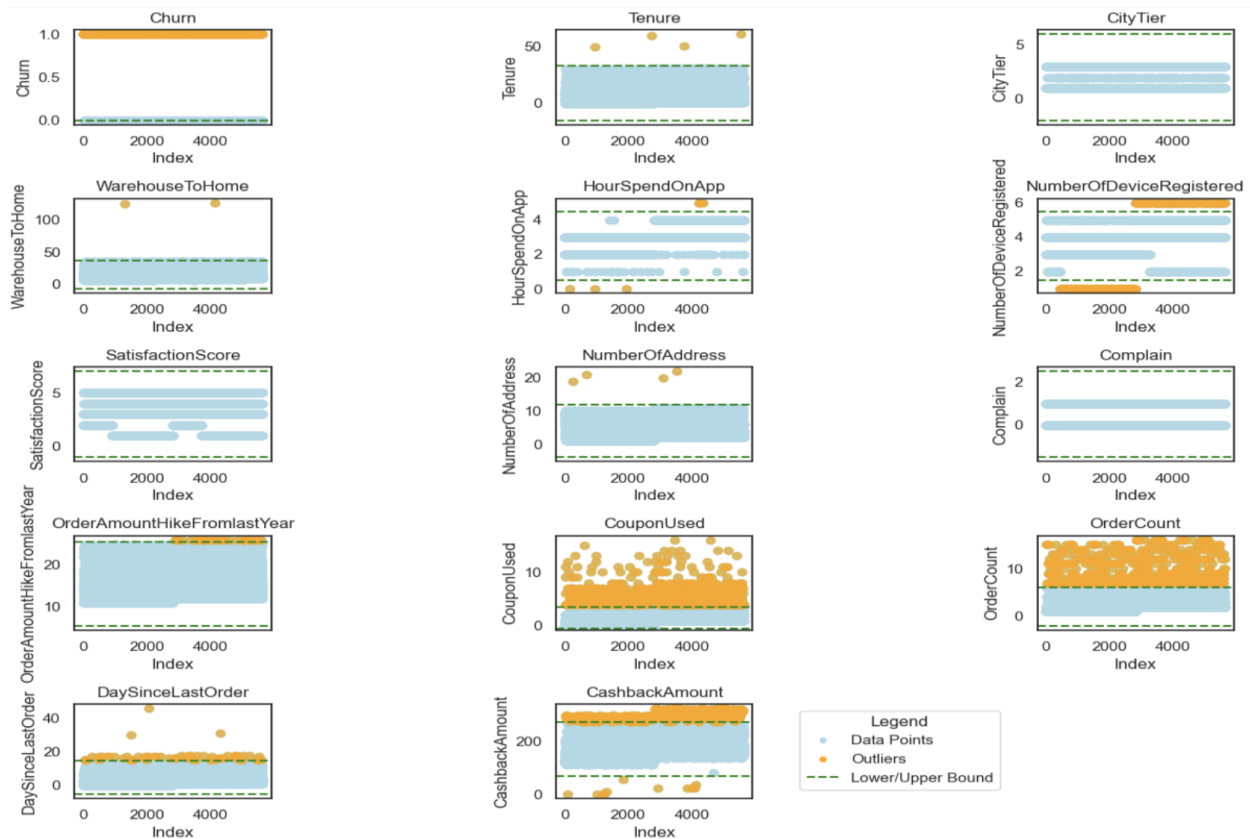


Fig 8: Box Plot for Outlier Detection

The assessment of outlier treatment, the data set had a total of 5630 records out of which 4682 records were of “Not Churned” and 948 were records of “Churned” which shows that there existed a huge bias in the two classes. This bias indicated how the cases of non-churning customers out stood in number the churning customers and made the modelling exercise quite constraining. But following those Outlier reduction techniques, the sample size dropped considerably such that Non – Churned cases diminished to the extent of three thousand two hundred and sixty-two while Churned cases reduced to seven hundred twenty-four. This Represents a decline of 30.33% of the majority class and 23.63% of the minority class. Such an alteration, and especially the one affecting

the minority class, was detrimental in that it raised fears over the possible difficulties in training the model as well as lowered its likely effectiveness in prediction.

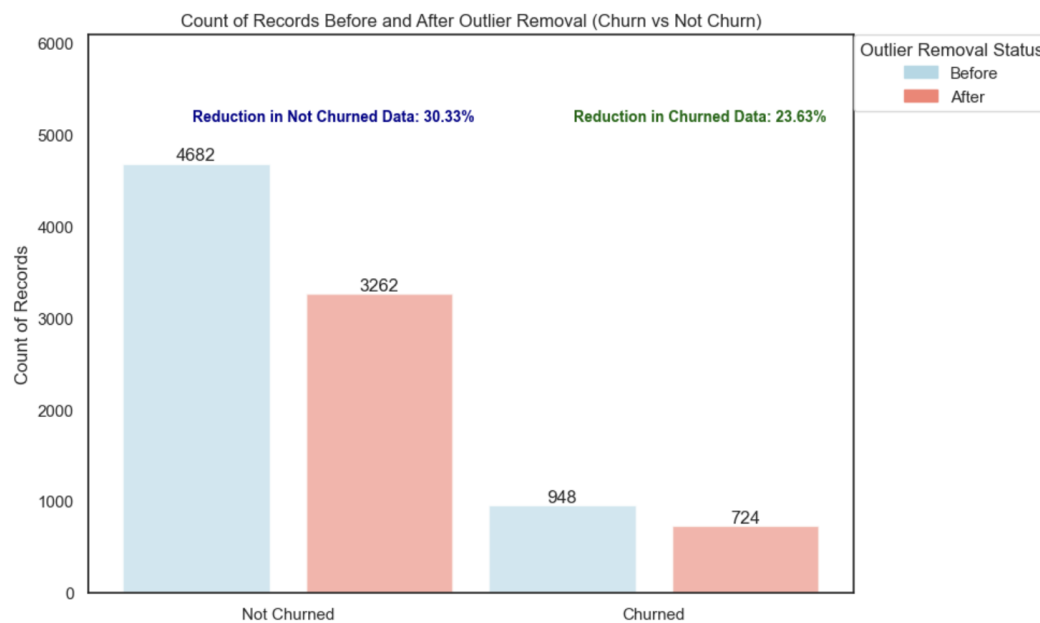


Fig 9: Distribution of Not churned and Churned Before and After the Outlier reduction

The elimination of meaningless data points also serves the purpose of enhancing the quality of the information, which it does, but also aggravates an existing problem of class imbalance which is very important for building a good model. This resulted in a negative impact, especially due to the absence of much needed information regarding the minority class, which is pivotal in solving the problem of customer churn. The end result could be that the model would be biased towards the majority class and customers at risk of churning would not be well identified. Based on these outcomes, it was decided not to use outlier filter technique, instead focus on retaining the original dataset. Therefore, the fewer poor cases, which in this case are also quite pertinent, would no longer be a wasted effort.

3.5 Feature Selection and Engineering

In order to enhance the performance of the model, feature selection and feature engineering were incorporated in this research with the aim of eliminating any repetition in the dataset.

In our dataset, the evaluation of the outlier removal process has shown some serious effect on the distribution of the dependent variable, which is customer churn status. The raw dataset in this case had 5630 records, of which 4682 were labelled as “Not Churned” and 948 as “Churned,” indicating a fair amount of underrepresentation of the less prevailing class as compared to the other. This imbalance implied that the number of “Not Churned” cases was more than sufficient compared to the “Churned,” and hence it was difficult to build an accurate model. However, post outlier reduction techniques, the total number of observations reduced drastically with “Not Churned” decreasing to 3262 and “Churned” records reducing to 704. This corresponds to a shrink of 30.33% in the majority class and 23.63% in the minority class. The decrease, particularly in the minority class, raised questions on the possibility of model training being biased and consequently reduced effectiveness of the model.

The removal of outliers irrespective of being a data cleansing measure, worsened an existent positive class problem, which is a key consideration in the building of a sound predictive model. This treatment results in absence of important data being removed, more so with the underrepresented group who all possess critical insights concerning customer churn. Such a scenario may create a risk of overfitting the model with the target class and this may lead to the risk of customer churn being overlooked. Considering these outcomes, retaining outlier observations appeared less critical than sustaining the greater integrity of the dataset. In this

manner, the proportions of the two categories are maintained, allowing more proper and useful functions to be performed with customer behaviours and churn rates analysis.

3.5.1 Feature Transformation

In this stage, it was important to convert the categorical variables into numerical formats compatible with machine learning models. One-hot encoding was performed on the PreferredLoginDevice, PreferredPaymentMode, and Gender variables, turning them into new binary columns. Such a transformation maintains the categorical structure but makes it usable for models which only take vector inputs (Tang *et al.*, 2020).

Also, feature scaling was carried out for the numerical columns to keep the data within a fixed range. Standardising was achieved using a StandardScaler which alters the features to have mean values of zero and variance equals to one. This is very important when using models where the distances for example the euclidean distance of the feature space is calculated because otherwise large scales of the feature analysis would dominate the analysis (de Amorim, Cavalcanti and Cruz, 2023).

3.5.2 Selection of Key Features

We are using the Variance Inflation Factor (VIF) which is a statistical method used to assess the level of multicollinearity among the analysed variables. This means that it describes how much the variance of a particular regression coefficient is increased when other predictor variables are intercorrelated.

In layman's terms, the VIF reveals if a particular predictor variable is presenting itself in a linear relationship with other predictor variables, which is usually not desirable. A predictor variable with a substantially high value of VIF means that it is collinear with one of the other predictors

within the model which may negatively affect the coefficient estimation and hence individual effect interpretation may not be very effective.

The VIF of an independent variable is expressed mathematically as shown below.

$$VIF = \frac{1}{1 - R^2}$$

where R^2 is the coefficient of determination obtained when the predictor is regressed on all other predictors. No correlation implies that $VIF = 1$. However, VIF above 10, exhibit concern with multicollinearity, which is likely to increase the standard errors and hence lower the effectiveness of the model. Excessive VIF may lead to the need for adapting the model and dropping one or more variables that are causing collinearity (Adnan, Ahmad and Adnan, 2006).

Based on our assessment of the Variance Inflation Factor (VIF) in relation to each feature included in our study, further complications of multicollinearity were identified. The feature CashbackAmount possessed also the highest value of VIF (66.92), indicating its very high degree of association with the remaining predictors. In addition to this, features like HourSpendOnApp (VIF 25.19) and OrderAmountHikeFromlastYear (VIF 19.25) presented high VIF values, showing the presence of redundant information. Thus, in order to improve the clarity and functionality of the model, the variable CashbackAmount was excluded from the dataset. The objective of this was to reduce the level of redundancy and lessen the impact of the multicollinearity problem within the study.

Feature	VIF
CashbackAmount	66.919575
HourSpendOnApp	25.191965
OrderAmountHikeFromlastYear	19.254500
NumberOfDeviceRegistered	16.825080
PreferredPaymentMode_Debit Card	11.147545
PreferredPaymentMode_Credit Card	8.295314
CityTier	6.109338
SatisfactionScore	5.892480

Fig 10: Initial VIF

After the exclusion of CashbackAmount, the VIF values of the remaining variables were evaluated again. The analysis on the updated data indicated that HourSpendOnApp was still characterised with a high VIF of 21.64, thus posing a serious concern regarding multicollinearity. Other features such as OrderAmountHikeFromlastYear with a VIF of 17.57 and NumberOfDeviceRegistered with a VIF of 15.25 were also still high. Consequently, we also excluded HourSpendOnApp from the dataset. After making these changes, all the other variables exclusive of the variable that was excluded had their VIFs cut off less than 10, which implied that the problem of multicollinearity was resolved.

Feature	VIF
PreferredPaymentMode_Debit Card	7.651634
CityTier	5.809863
PreferredPaymentMode_Credit Card	5.781391
SatisfactionScore	5.593734
WarehouseToHome	4.293080

Fig 11: Final VIF

Therefore, further analysis was possible at this point, as it was clear that the features included in the analysis of variance would be more useful and easy to understand in terms of the model constructed.

3.6 Model Development

3.6.1 Data Preparation and Sampling

The initial phase in developing a model is to partition the dataset into independent variables (X) and a dependent variable (y). In order to form the feature set X, all the columns except for the “Churn” column in the dataset are retained while the column “Churn” is dropped. The column ‘Churn’ serves as the dependent variable (y) as it indicates the status of the customer in relation to whether or not they have idle status. Considering the considerable class distribution skew contained in the dataset where the non-churned customers are greater compared to their churned counterparts, it is of great importance that this class imbalance is appropriately dealt with in order to avoid the models getting overfitted toward the dominant class. In order to resolve this problem, SMOTE (Synthetic Minority Over-sampling Technique) was employed. By creating new samples

in the ‘minority’ class, the proposed SMOTE prevents the class overlap and improves the class ratio evenly allowing good training performance for both classes.

Two functions are created to assist in this process. The first one is called `split_data` and it accepts the feature set `X` and the variable `y` as arguments along with the corresponding data in order to perform a division of the data into training and testing levels, with 20% of the data set aside for testing purposes. This enables models to train on sufficient data whilst retaining some for performance evaluation. The second function, `apply_smote`, is directed towards addressing class imbalance in the training dataset by offering synthetic sample upsizing technique, which results in the training dataset containing a more equal sized class distribution. This will be achieved based on the assumption that training the model in this case with SMOTE will improve the model's chances for correctly predicting the churned customers in the model. Thus, the overall accuracy of the prediction will increase.

3.6.2 Comparative Analysis of Model Performance: Before and After SMOTE

In order to evaluate the effectiveness of SMOTE with respect to model performance, we performed the comparative analysis of three machine learning models Logistic Regression, Random Forest, and XGBoost respectively: before and after using SMOTE. The results show how each model responds to adjusting the classes of data and its implications on customer churn predictions.

On the other hand, Logistic Regression produced an accuracy of 90.76% pre SMOTE application, which is rather a good figure in total classification perspective. Churn cases, however, proved more difficult for the model – apparent from a recall of only 54.05% for the metric. In this case, that means nearly half of the churners were missed by the model. Moreover, the F1 score of 0.6579 indicates the extent of this disparity between precision and recall, as Logistic Regression more

often than not, predicted only non-churn cases due to the class imbalance. After setting out SMOTE, there was a dramatic change in the performance of the model. The accuracy of Logistic Regression reduced drastically to 80.46% with a precision drop of 44.74%, which means that the model began to have more false positive predictions. There was however a notable change in recall, which shot up to 80.54%, showing that the model became much more capable of predicting churn cases. The trade-off here was in turn a lower F1 score of 0.5753, which indicated that while the model was able to predict a lot more churn cases, it did so at the cost of precision, thus having a greater likelihood of false positive predictions than was acceptable. This change shows that Logistic Regression as a linear classifier is prone to overfitting when synthetic samples are added due to the modelled behaviour shifts in the decision boundary and thereby increases in classification errors.

Model	AUC (After SMOTE)	AUC (Before SMOTE)	Accuracy (After SMOTE)	Accuracy (Before SMOTE)	F1 Score (After SMOTE)	F1 Score (Before SMOTE)	Precision (After SMOTE)	Precision (Before SMOTE)	Recall (After SMOTE)	Recall (Before SMOTE)
Logistic Regression	0.889048	0.891990	0.804618	0.907638	0.575290	0.657895	0.447447	0.840336	0.805405	0.540541
Random Forest	0.986642	0.987506	0.969805	0.973357	0.903955	0.912281	0.946746	0.993631	0.864865	0.843243
XGBoost	0.982951	0.984134	0.979574	0.976909	0.935933	0.927778	0.965517	0.954286	0.908108	0.902703

Fig 12: Comparison of model evaluation metrics before and after SMOTE

The performance of Random Forest was in contrast to Logistic Regression. Before SMOTE was applied to the model, Random Forest obtained 97.08 % in terms of accuracy, 96.91% in precision and 84.86% in recall which are impressive values indeed. These metrics speak volumes of the model's capacity to bear on churn prediction such that a majority of the carcinoma cases are able to be diagnosed without compromising on generality. When it came to performance after SMOTE

was executed, it was still strong, although there was a slight reduction in accuracy to 96.98 % for Random Forest. Its precision decreased slightly down to 94.67%, but recall was up at 86.49%, this quite logically means that a greater number of churn cases were identified while reducing the number of false alarms to an acceptable level. The small precision drop might be explained by SMOTE and its synthetic samples, but in general, such a balanced dataset was dealt with quite well by Random Forest, with no considerable drop in performance. Such performance maintenance both before and after the application of SMOTE underscores Random Forest's reliability in predicting churn, even when presented with a relatively balanced dataset.

Before the application of SMOTE, XGBoost performed extremely well, resulting in 97.69% accuracy, 95.43% precision, and 90.27% recall. These figures, coupled with the results from the Logistic Regression and Random Forest models, showed that XGBoost was quite useful in identifying churners even with the existing class imbalance. The recall of XGBoost surpassed that of both Logistic Regression and Random Forest models. After SMOTE was applied, XGBoost still performed Santiago-level, further increasing its accuracy to 97.96%, precision to 96.55%, and recall to 90.81%. The growth in precision and recall following SMOTE also shows XGBoost's capability to utilise a balanced dataset to boost its predictive power. On the contrary to Logistic Regression, XGBoost withstood the constraint of synthetic samples and hence overfitting, and its remarkable performance all-round depicts its strength as a classifier, even for highly imbalanced situations.

Throughout the modelling comparison, it was evident that Logistic Regression fluctuate the most pre and post application of SMOTE, where its capacity to identify churn cases improved significantly at the expense of overall precision and accuracy. However, measures of effectiveness

pertaining to the performance of Random Forest and XGBoost incorporated before and after SMOTE remained high with few variations in their measures. This implies that the use of SMOTE does not considerably affect the performance of class encoders such as Random Forest and XGBoost. The high recall values associated with Random Forest and XGBoost post SMOTE, along with their precision values, indicate that these models are very useful for churn prediction problems, especially those where identifying the churners is of utmost importance. The strength of this allows the use of Random Forest and XGBoost in carrying out Churn predictive Modelling, on the contrary to Logistic Regression which may not be effective due to the nature of the class imbalance problem and reliance on synthetic data.

3.6.3 Cross-Validation to Address Overfitting

To further understand the stability and reliability of each model, cross-validation was conducted on the Logistic Regression, Random Forest, and XGBoost modelling approaches. Cross-validation is used to limit overfitting by dividing the whole dataset into a number of its divisions or folds and training an algorithm on some other divisions at every other time. Thus, the given method makes it possible to define more accurately the generalisation of every considered model (Berrar, 2019). Applying Logistic Regression, the mean AUC for the 5 folds was 0.8863, with AUC 0.8799 in the lowest and 0.8937 in the highest fold. These numbers prove that although Logistic Regression is quite useful, it is not advanced enough to outdo the performance of tree based models when it comes to targeting churners. Given their relatively lower AUC scores, it may be difficult to imagine Logistic Regression as the appropriate model to deploy when seeking to classify churners versus non-churners, particularly when there is a class imbalance.

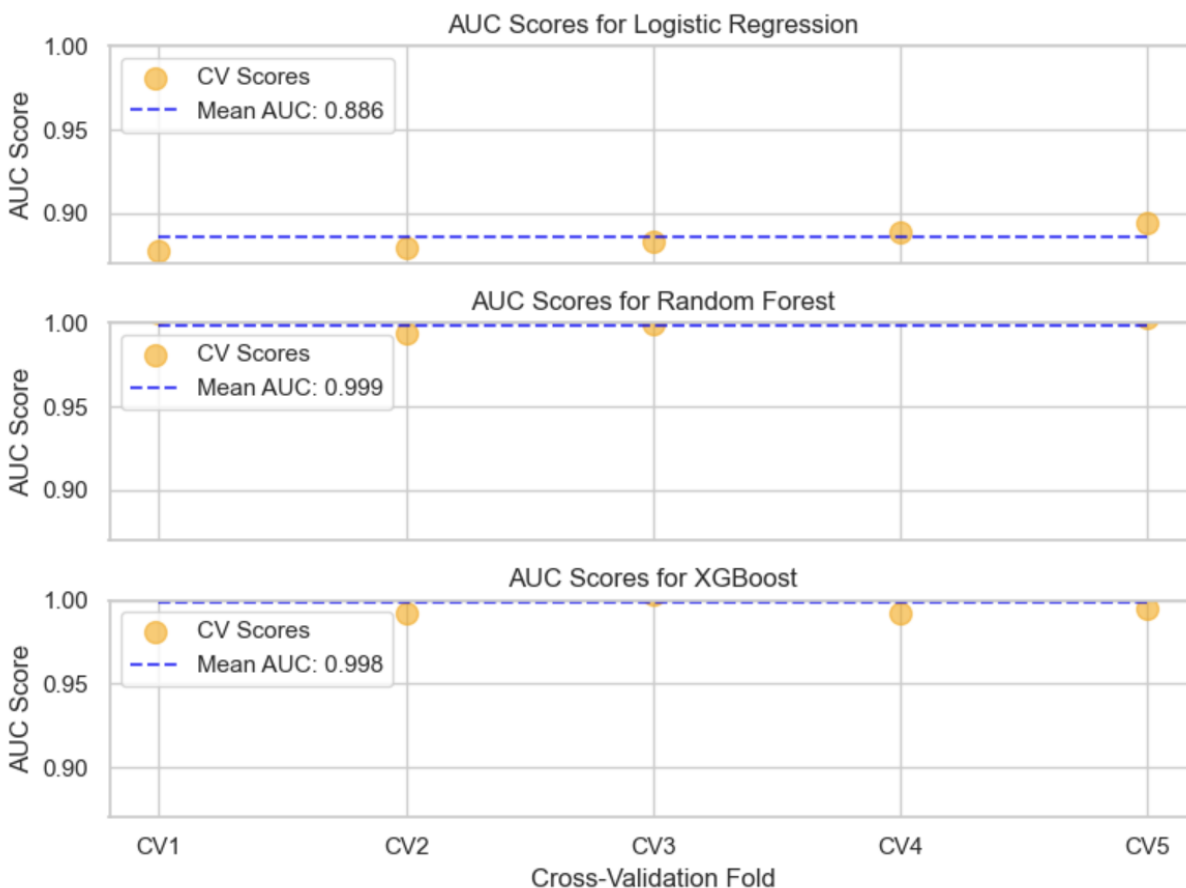


Fig 13: Comparison of AUC score of models Before and After Cross Validation

During the cross-validation, Random Forest performed remarkably well obtaining a mean AUC score of 0.9987 with no fold less than 0.9977 and no fold more than 0.9998. Such a high degree of stability illustrates Random Forest's superiority over other models when identifying churners and non-churners. Thus, customer churn can be predicted efficiently with very little chances of overfitting this model. Quite to the same extent, XGBoost also performed with a high mean AUC of 0.9983 while the fold AUC scores ranged from 0.9962 to 0.9994. The results here also show XGBoost to be effective as a classifier which is consistent with the results obtained before and after application of SMOTE. The performance of XGBoost and its ability to achieve consistent

AUC in all folds in support of his performance in the presence of class imbalance.

3.7 Ensemble Methods for Enhanced Model Performance

Ensemble methods were also investigated to improve predictive performance owing to the commendable performance of Random Forest and XGBoost. Two approaches to ensemble learning were applied: the Voting Classifier and Stacking Classifier.

The Voting Classifier Aggregates Random Forest and XGBoost models by using soft voting.

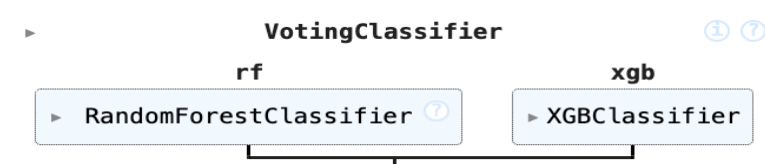


Fig 14: Voting Classifier

This means that the predicted probabilities of each class will be taken into account. After training the base model on SMOTE generated data, the Voting Classifier obtained the mean AUC of 0.9973 which is remarkable Lin. Such a high AUC score indicates the ability of the model to differentiate between churners and non-churners incorporating individual strengths of Random Forest and XGboost.

A Stacking Classifier was also created where Random Forest and XGBoost were the base models and lastly Logistic Regression was the top model.

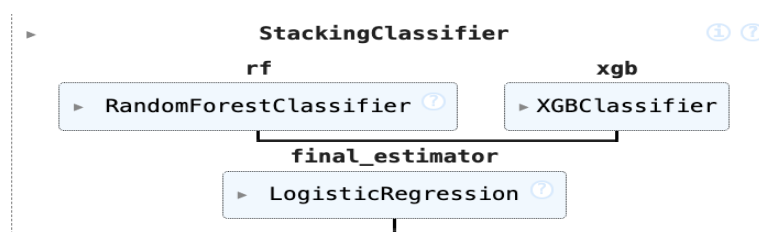


Fig 15: Stacking Classifier

The purpose of this classifier is to learn from the predictions of the base models in order to produce better final predictions. A mean AUC score of 0.9971 was also achieved by the Stacking Classifier indicating its considerable prediction performance capabilities. Stacking Classifier in regard of AUC although is lower compared to Voting Classifier is still a favourable candidate for churn prediction.

To sum up, both ensemble approaches improved the models, with a slight advantage of the Voting Classifier over the Stacking Classifier. Considering the small discrepancy in their outputs, both methods provide a dependable way for enhancing the predictive performance in customer churn analysis, nevertheless, the Voting Classifier tends to be preferred in this case as it has a marginally higher AUC score.

3.8 Ensemble Model of Voting Classifier

3.8.1 Model Training and Testing

The Voting Classifier has been designed to capitalise on the predictive power of Random Forest and XGBoost by implementing soft voting. Unlike hard voting, soft voting uses the predicted probabilities pertaining to class membership by these models, thus allowing the Voting Classifier to take advantage of the mutually exclusive features of the two tree models. With this technique, the model has a better chance of making correct predictions. This is accomplished using the balanced dataset created with the SMOTE step where the minority class of churners is better populated. Then the model is explained in detail and tested using cross-validation which is an effective technique for assessing predictive accuracy of a model built on one dataset and tested on unseen data by providing estimates across various groups of data to avoid overfitting and enhance the applicability of the findings to other fresh datasets.

3.8.2 Evaluation Metrics

The performance of the voting classifier is evaluated based on a number of metrics including, but not limited to, accurate rate, precision rate, recall rate, F1 score and AUC. The AUC score, recorded to be 0.9973, indicates how well the model is capable of distinguishing churners from non-churners. The area under the curve (AUC) is high showing that the model is effective in per class categorization hence can be used in predicting customer churn with high effectiveness. This score is consistent with the performance exhibited by the two base models Random Forest and XGBoost which suggests the efficacy of the Voting Classifier in reaping the benefits of both models.

Confusion Matrix

The Confusion Matrix of the Voting Classifier classifies the performance of the model in accordance with True Positives, True Negatives, False Positives, and False Negatives.

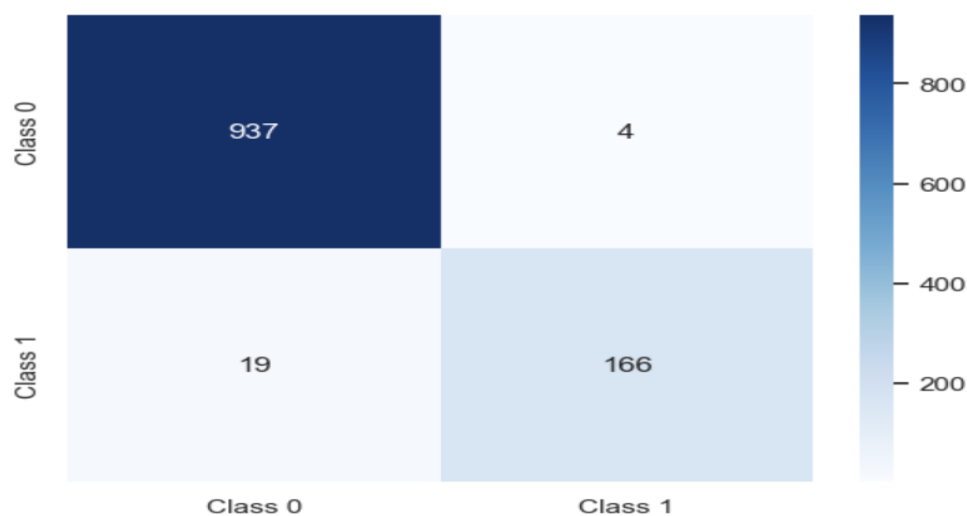


Fig 16: Confusion Matrix

This representation is important to visualise the consequences of the predictions made by the model. For instance, a high true positive rate (treating churners correctly) suggests that the model is effective in spotting potential churners, thus aiding the company to concentrate its retention strategies even more. On the other hand, a low false positive rate implies that a smaller proportion of loyal customers are wrongly considered as at risk of churning thus carrying out unnecessary measures on them is avoided.

Accuracy and Precision

The Voting Classifier achieves a crystal-clear accuracy of 98%, which implies that making mistakes is nearly impossible with the model. This level of accuracy is of great importance for organisations that need to make exact predictions as to when a customer will churn in order to develop appropriate intervention measures. The precision which tells how many of the total churn risks predicted were correct is also noteworthy at 98%. This means that if the model says the customer will churn, there is a 98% chance it is correct. High precision makes it possible to focus retention efforts on areas where they are needed and not to waste marketing on people who are not in danger of leaving.

Recall and F1-Score

As we have previously mentioned, recall, which pertains to the percentage of actual churners the model is able to correctly identify, is one of the most important measures in predicting churn. The Voting Classifier boasts a churner recall rate of 90% which means that it correctly classifies 90%

of the customers who have a chance to churn. While this is an advantage in many instances where high recall is sought in order to capture every at-risk customer, it however indicates that 10% of the churners, who could possibly be identified, lie on the other side of threshold suggesting room for enhancement. For every other group of whom we expect them not to churn, the classifier attains a recall score of one which means all the non-churning loyal customers are correctly identified with no risk of them being labelled as churners.

The F1-score, which assesses the degree of precision and recall that is utilised for the two classes, is commendable high. This measure is important in the context of predicting customer churn since there is a need to minimise the risk of classifying a customer who is at most risk (high recall) into a low risk (high precision) group. A high F1-score in the presence of churn cases also means that the model is good at identifying sets of churners and is also cautious in making such claims which makes it appropriate for enhancing customer retention initiatives.

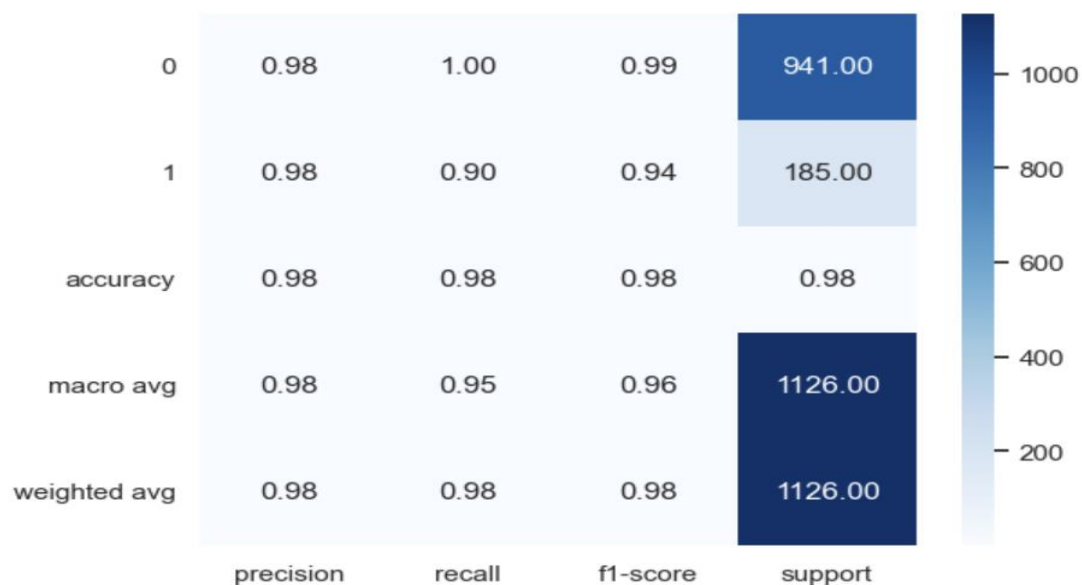


Fig 17: Evaluation Metrics for Voting Classifier

All in all, a Voting classifier with an impressive accuracy of 98% implies that most of its predictions are correct, allowing companies to reasonably spend resources on their customer retention initiatives. The remaining high F1-scores show how well the model achieves the objective of precision and recall, which is critical in solving the churn problem which is similar to the churn prediction model using ensemble models (Wang, Nguyen and Nguyen, 2020). In other words, these findings seem to reassure businesses that the model can successfully be employed to enhance customer retention and minimise customer loss. However, this should not make the businesses complacent and they should keep in mind that there is always a small percentage of customers who are at the fringes and hence need other engagement approaches to win them back. This knowledge will subsequently help organise customer retention efforts more efficiently, and as a result improve customer satisfaction and loyalty within the firm.

4. Result

An advanced structure for analysing customer churn in e-commerce featured a machine-learning approach, which was used to determine the important factors that can help in retaining customers. As the level of rivalry grows in the online world, it has become very important for the growing e-commerce industry to learn in detail what makes customers loyal to a service or product. Customer churn is defined as the percentage of customers who stop doing business with a company and it is one of the most problematic issues that can hinder the revenue and growth of any business. To help out with the users' retention strategies, sophisticated drivers of decision-making processes were uncovered.

After conducting a thorough analysis of several types of customer data, the study identified a number of major attributes that help to explain the relationship between customers and their

propensity to leave the service. These attributes indicate potential opportunities that the e-commerce organisations should exploit to enhance customer retention and satisfaction. Recognizing these predictors gives companies the chance to design specific measures to deal with the problem of customer defecting to another company.

Top 10 combined important features:

	Feature	Mean_Importance
index		
1	Tenure	0.203630
2	Complain	0.099785
3	DaySinceLastOrder	0.063714
4	NumberOfAddress	0.056803
5	CouponUsed	0.049733
6	SatisfactionScore	0.049309
7	PreferredOrderCat_Laptop & Accessory	0.048240
8	MaritalStatus_Married	0.042189
9	WarehouseToHome	0.041157
10	CityTier	0.038346

Fig 18: Top ten important features

The insights elucidated in this study also offer a baseline on the extent and impact of various factors on customer attrition in the e-commerce sector allowing for the proper formulation of strategies geared towards the retention of customers.

5. Discussion

Interpretation of Result

In the context of e-commerce, understanding the reasons behind customer churn presents some complex issues on the retention and the loss of customers. As such, this study employs machine learning algorithms to determine the key predictors of churn which helps businesses in informing the most suitable retention strategies to think of based on determinants that are evidence-based.

The characteristic of Duration that stood out as the most important predictor of churn implies that the longer a customer has been associated with a brand, the more resistant they will be to leaving. The declining attitude of customers' behaviour and the growing customer tenures have reinforced the idea that there is a need for businesses to create and maintain long term relationships with their customers. It has been established that these kinds of programs reduce churn rate because they help establish and maintain encouraging relationships that are tied to frequent and repeat buying behaviours (COŞER *et al.*, 2020).

While aspects like product quality or pricing may foster customer loyalty, it is important to note that dissatisfaction and more specifically, complaining behaviour is pivotal in ensuring customers stay within a given business. Customers who have issues that are not resolved will most probably leave. Thus, it is essential that the company applies some efficiency in the customer service practices. According to Knox and Van Oest (2014), the encouragement of customer relations, especially complaint handling is not only a necessary step for conserving the existing customer base. It can also be used to create a more positive image of the customers which in turn enhances their loyalty.

The metric DaySinceLastOrder expresses the involvement of customers in the business activity and implies the dependence of the churn on the time since the last order of a specific customer within a given time period. This means that companies should enhance contact with such customers who haven't placed orders recently, for example, by sending notifications or special custom made offers. In order to win back such customers, it is critical to keep communication channels open and prompt them back to the platform (Mokadikwa, 2008).

The metric of NumberOfAddress highlights that customers who have multiple shipping addresses

are more active and tend to stick around for much longer. This shows trust and attachment to the e-commerce platform from the customer, which implies that this kind of behaviour can be analysed and this can be used by the businesses in adjusting their marketing campaigns effectively. Providing offers customised to the specific locations or taste may lead to higher levels of customer retention (Zorn, Jarvis and Bellman, 2010).

The observation associated with `CouponUsed` suggests that although discounts are typically useful in bringing in customers who are more apprehensive towards price, businesses must learn to strike a balance in these techniques so as not to destroy the brand equity. This clearly points to the requirement of devising tactical promotional measures which seeks to enhance loyalty and at the same time, those who are satisfied with the price (Seo, Choi and Yoo, 2022).

It is certain that `SatisfactionScore` is related to retention in the sense that people with higher satisfaction levels are less likely to churn. That is why, in turn, every company should strive to enhance service level and client experience on a daily basis (Craucamp, 2012). Without doubt, one way to achieve this is collecting data to regularly measure business.

The strategic asset of `PreferredOrderCategory` focuses on recognizing customer preferences which are vital for the successful implementation of marketing strategies. Customers that have an interest in specific categories are more likely to be drawn in and retained due to specific promotions made to them (Park and Han, 2013).

The demographic facet represented by `MaritalStatus` underlines however that it is always possible to influence loyalty through various marketing strategies depending on the demographics of the customers. It is precisely such nuances that allow businesses to implement more sophisticated designs of the campaigns and address them to particular customer segments (Abiad and Ionescu,

2020)

As depicted in WarehouseToHome, having developed strategies for effective delivery mechanisms is paramount in ensuring customer satisfaction. As such, deliveries are made on time within a reliability scale which influences those businesses to streamline their logistics so as to foster customer loyalty enhancement (Vakulenko *et al.*, 2022)

The CityTier functionality stresses the importance of geographic segmentation in time and place marketing. In brief, different groups of customers based on their location have different requirements through which better engagement and retention by the businesses can be achieved (De la Llave, López and Angulo, 2019)

This means that from an e-commerce perspective, multiple areas of concern may lead to customer churn. However, the nuances of why customers leave help organisations to devise suitable measures during the appropriate timing and target the desired audience with the right appeal.

Implications for Customer Retention Strategies

The results of the feature importance analysis can be crucial in formulating e-commerce customer retention strategies. E-commerce businesses also need to focus on designing good strategies for their customers who are churned. For example, the company can implement a loyalty program whereby customers who have been with the company for a long time are rewarded according to their time of stay in the organisation. This enables the business to appreciate the significance of the customers' loyalty and reward them so that their spending is sustained.

Moreover, the resolution of customer complaints should not miss any retention strategies because

it is a priority. For example, explaining what will be done to avoid a situation from getting worse using communication strategies is very helpful. It refrains customers from having the worst experience and wishing to leave the customer base. There is evidence in literature that suggests that when customers believe that they are valued and are listened to, they are likely to stick to a brand even after it fails them (Day, 1999).

Besides, companies must be proactive in connecting with clients that appear to be withdrawing from the business, as illustrated by the high metric of `DaySinceLastOrder`. To spark their interest and motivate them to make purchases again, re-engagement campaigns, customised emails, and special deals can be very helpful.

Practical Application of the Churn Prediction Model

Thanks to the churn prediction model, e-commerce businesses can now take necessary measures to retain customers. For instance, churn predictors strive to identify those reasons which are salient to certain segments of consumers and develop strategies targeting those reasons in those consumers. It abandons the reactive strategy and resolves the issue long before it becomes dramatical- ere the customers are lost.

In this way, even a complaints model may contribute to business performance. For instance, the churn prediction model may assist in identifying a segment of customers whose behavioural singularity is likely to lead to their dissatisfaction to the firm and then such measures may be employed, for example, other pre-emptive actions like improving customer service skills or making their complaint resolution process easier. This kind of reasoning also leads to increased retention through building a happy customer experience where all issues have been pre-empted

(Knox and Van Oest, 2014).

Furthermore, examining customers' actions with the help of feature importance allows businesses to determine the effectiveness of their advertising efforts. For example, knowing that CouponUsed is positively associated with retention helps in devising strategies that may include low prices to encourage such customers back. Such campaigns can be assisted with past purchase information so that they are relevant to the target market for effectiveness (Seo, Choi and Yoo, 2022).

6. Limitations Of The Study

Even though this research in many ways is very helpful, one has to keep in mind several limitations. One such limitation is the dependence on a single source of data which might exclude some factors pertinent to customer behaviour. The analysis is restricted to the characteristics present in the model at hand and does not extend to factors such as trends in the external environment, the economy or competition, which may be relevant to customer churn in the field of e-commerce. An expanded data set with all these factors included would lead to one better understanding of customer behaviour.

While preparing machine learning model, hyperparameters could be explored to improve the model's performance.

In addition, the current research is overly concerned with numerical data analysis, which can ignore certain aspects of customer experience. How understanding the emotions and motivations of customers with respect to their behaviour can help in understanding the dynamics of churn better. Looking further, already available data could include more qualitative aspects such as customers' perspectives drawn from interviews or discussion groups to shed more light on causes of churn.

7. Future Research Directions

In future studies, it would be beneficial to assess churn with additional variables, such as external market forces and economic factors. This would help in appreciating the intricacies of customer behaviour in the context of electronic commerce. In addition, it would also be important to look in this context at different AI or machine learning models and their role in the customer retention management, as these aspects are equally important for the ongoing fight against the churn.

Further research may examine how effective different retention techniques are in regard to clients' categories, which would in turn foster more targeted and effective retention strategies towards reducing churn and instead of focusing on particular customer in the implementation part, could build more robust system to analyse group of customers to get insights into retention methods in a group level. Longitudinal studies would also be relevant since they would focus on how customer behaviour changes over a certain period of time which would assist in explaining the trends in the e-commerce scenario as well as customer retention techniques.

8. Conclusion

Overall, the features like Tenure, complaints, and the duration since the last order are some of the key factors that determine customer churn. The findings of Wang, Nguyen, and Nguyen (2020) are corroborated by a Ensemble model Voting Classifier that combines both Random Forest and XGBoost classifiers, which yielded the highest performance reached recall (90% and above) on churn predictions. Customer churn can be reduced through measures such as loyalty programs, feedback surveys, resolving issues before customers complain, and providing better deals for those who do not order regularly.

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10. Appendices

Ethics Review Form:

Preview of Web App for implementation:

Customer Churn Prediction

Customer Details

Tenure(months)

Satisfaction Score

Complain

Days Since Last Order

Coupons Used

Distance(KM) between Warehouse and Home

Number of Addresses

Order Count

City Tier (1 to 3)

Marital Status

Preferred Login Device

Preferred Payment Mode

Gender

Preferred Order Category

Predict Churn

Prediction Result

Churn Status: Likely to Churn

Churn Probability: 84.16%

User Input:

- Tenure: 0
- SatisfactionScore: 2
- Complain: No
- DaySinceLastOrder: 120
- CouponUsed: 3
- WarehouseToHome: 27
- NumberOfAddress: 7
- OrderCount: 6
- CityTier: 1
- MaritalStatus: Single
- PreferredLoginDevice: Mobile Phone
- PreferredPaymentMode: Debit Card
- Gender: Female
- PreferedOrderCat: Laptop & Accessory

Suggestions To Avoid The Churn Of This Customer:

- Engage the customer through targeted loyalty programs to increase their tenure.
- Conduct a survey to gather feedback and improve customer satisfaction.
- Implement a proactive outreach strategy for customers with complaints to resolve issues quickly.
- Send reminders or incentives to encourage more frequent orders from customers.
- Evaluate delivery times and explore options to enhance the efficiency of the delivery process.
- Offer exclusive deals for customers in Tier 1 cities to attract them.

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GitHub Repository for the Code: https://github.com/aktrikx/mp_cp_ak